

• Easy #14 • Longest Common Prefix

▲ Problem Statement → Given an array of strings, find the longest common prefix. If there is none, return "".

▲ Main Solution → Use the first string as a reference. Check each of its characters at the same position in all other strings.

If all match, add the character to "prefix". If not, return "prefix".

▲ Brute Force - Compare every character of every string from the beginning until a mismatch or the end of a string is found.

▲ Algorithm → 1. Create an empty string "prefix".

2. Loop through each character of the first string.

3. Check the same position in all other strings.

4. If a string ends or characters don't match, return "prefix".

5. Otherwise, add the character to "prefix".

6. Return "prefix".

▲ Important Points →

- Use the first string as the reference

- Compare characters at the same position

- Stop when a character differs or a string ends.

- "prefix += current" adds a matching character.

▲ Complexity → Time → $O(n \times m)$

Space → $O(m)$

n :- No. of strings

m :- Length of the shortest / common prefix.